CorDuro Housing: Survival curve analysis on B2B conversion sales rate[1]

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Abstract:

This paper investigates the perceived underperformance of B2B sales conversions within a furnished housing provider. Initial executive estimates hypothesized a sub-10% win rate, raising concerns of significant misalignment with standard industry benchmarks. To correct for the statistical noise introduced by right-censored, open pipeline data, this study applies Kaplan-Meier survival curve analysis to evaluate the true sales lifecycle and conversion baseline from initial internal approval to final contract execution. The analysis refutes the initial hypothesis, revealing actual conversion rates of 30% for existing partners and 13% for new partners, with 95% of successful deals closing within an industry-standard 163-day window. Ultimately, the survival models demonstrate that execution velocity and win rates are structurally sound once a contract is actively drafted. Instead, the analysis identifies a severe upstream operational bottleneck, demonstrating that 51% of approved deals are abandoned prior to formal contract generation. These findings redirect organizational focus away from late-stage sales execution toward resolving early-stage pipeline friction.

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[1] Note: Absolute pipeline volumes and specific organizational identifiers have been anonymized to protect confidential business data. Proportional conversion rates, lifecycles, and survival thresholds remain mathematically accurate.

The Business Model

CorDuro Housing (CDH) is a furnished housing solution provider that sells their product both ways: to Members and Partners. On the Partners side, we acquire management and lease rights (without being property managers) by signing a Deal for a certain number of units in a Property. We then install furniture according to our brand standards, and rent them out to our Members for short or long periods. The product works in two ways: we fill out Properties' units, and facilitate the leasing process to Members with simple, four pages leases.

CDH then earns money by entering into a Revenue Share Agreement with the Partners. By splitting the money in a certain ratio (varying according to performance projections and negotiations), we give the Partners a the majority of the revenue generated. This revenue tends to be similar to their Multifamily Equivalent, given the markup price we charge for providing a furnished apartment. The type of market we target are Class A and Class B buildings across the United States of America, in areas where our Interstellar Capital (IC) determines we would perform well.

In September 2025, the IC was established to direct the Sales department so we only sign Deals beneficial to CDH and in areas with easier access. Using forecasting metrics and information from sites such as CoStar, the IC reviews the location, viability, building maintenance, and other variables important to the business.

One of the main methods to verify investment opportunities is using an economic model called a Forecast. Each of these is tied to a specific Deal, not Property. A Property can have multiple Deals, each corresponding to a Forecast, catalogued as New Deals or Expansion Deals. These Forecasts contain the most relevant information for the sale, which is the number of units per bedroom disposition, along with their rent, and a calculation of the expected revenue over the term of the Deal (which varies from one, three, or five years). The Forecasts also have estimated occupancy rates, comparisons to Multifamily Equivalents, revenue share, and other information.

Note, that Deals can also be Standard or Lease Ups, where the latter is a Deal offered to buildings in construction to help them fill out their units faster. We usually acquire the rights for 120 units, then scale back as their occupancy rates raise, ending up with 30 units for ourselves.

The Forecast is generated by each salesperson using the information available on the Partner's website at the time of creation. Each salesperson must calculate the average rent cost for each of the unit types that CDH is interested in acquiring (Studios, 1-Bedroom units, 2-Bedroom units, and 3-Bedroom units) at that Property using the listed rent for the units at the Property's website. After adding these values, the Forecast calculates a blended rent by weighing the average rents per unit type (weighted average). This Forecast will be used by the salesperson as a sales tool to get a potential Partner on board with CDH's program.

First, the Forecast and the Property must be analyzed by the IC (sometimes, with an on-site tour). If it is approved, the pitch moves into the creation of an Order Form which is approved by the Deal Desk (following the previous IC approval). Then, the salesperson negotiates. Redlines may come back from the Partner over the entire Services Agreement governing the Order Form. The Deal Desk will reject or approve the requests according to a Playbook which determines who must approve certain requests, which go from the President himself, the VP of Development, General Counsel, or other people from the IC. If the negotiations are successful, the Partner will sign. Then, the field operations team and home coordinators will step in to govern the relationship with the Partner and make the required installations.

Sales Lifecycle and Conversion Rate

Since July 2025 and up until April 2026, there have been 6,273 Forecast review requests to the IC. The IC does fifteen-minute daily meetings to review these requests. The President of the company estimates that less than 10% of the Deals that have been approved by the IC get through the finish line and are signed. The IC has a 92% approval rating on the Forecast requests. A rough estimate shows that less than 192 Deals, that have gone through the IC, have been signed.

This gap raises concern: a lot of resources have been allocated to the IC, and most of those efforts are wasted. If it's true that less than 10% of Deals are signed, ten CDH's win rate is much lower than that of comparable industries. According to HubSpot's baseline performance metrics, average B2B sales win rates sit at 21% across all pipeline opportunities, rising to a 29% close rate for deals that reach the qualified, late-stage pipeline.[2] Salesmotion states that existing relationships should see win rates between 30% and 40%, while cold outreaches show win rates between 10% and 18%. Specifically, they set the desired zone for most B2B businesses between 20% and 35%.[3]

Why is CDH's win rate presumed to be so low? What is the true sales lifecycle and baseline conversion rate when we account for right-censored, open pipeline deals? To answer these questions, we will use a Kaplan-Meier survival curve applied to CDH's sales pipeline, starting with the Deals found in Czerka and, after, compared to the Interstellar Capital's approval log.

[2] "Want to create a sales plan? Let me show you how [+8 sales plan examples]." Jay Fuchs, 2026. HubSpot. https://blog.hubspot.com/sales/ultimate-guide-creating-sales-plan#:~:text=Forecast%20accuracy%20improves%20when%20teams,intervene%20before%20the%20quarter%20ends.

[3] "Sales Win Rate: How to Calculate and Benchmark in 2026." Semir Jahic, 2026. Salesmotion. https://salesmotion.io/blog/sales-win-rate-benchmarks-2026

Model proposal

We propose a Kaplan-Meier survival curve on the deals to solve two issues. First, the Customer Relationship Management (CRM) data is not consistently purged. For that reason, any unsigned deal will introduce noise. The second issue is that a simple ratio of signed-to-approved deals does not consider the sales lifecycle or cohort maturity. Any deal approved at the end of the observed period will inflate the denominator of a simple division.

Upon reviewing the available data, we came to the following information. Since November 1st, 2025, the IC has reviewed 4,902 Deals/Forecasts, of which 4,455 were approved (roughly 91%). In Czerka, the platform used by CorDuro since November, 2,208 Agreements were created for the purpose of signing new Units to CorDuro's portfolio. That is, 49% of Forecasts approved ended in a contract being generated.

As noted previously, Forecasts approved by the IC in the latter months of the observation window (e.g., April) are right-censored; they have not yet had adequate time to traverse the pipeline. To establish an accurate conversion rate, it is necessary to measure the full lifecycle duration, from initial IC approval to final signature, to determine the true probability of conversion over time.

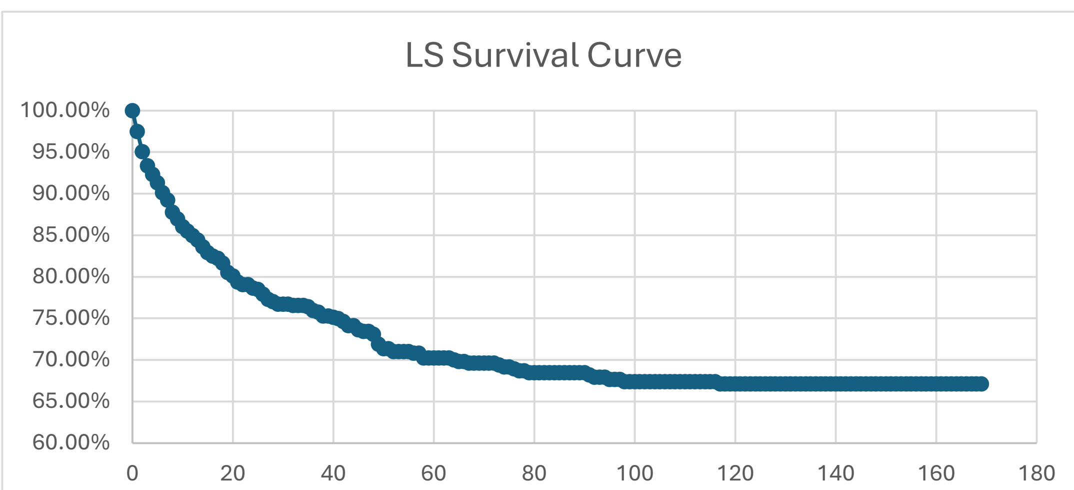
Kaplan-Meier Survival Curves

How long does it take for an Agreement to be signed? What is the cutoff point, or threshold, where existing Agreements will no longer be signed? To answer these questions, we propose using two survival curves. The first measures the lifecycle of a deal from its creation in Czerka until the cutoff date of April 23rd, 2026. The second curve measures the lifecycle from the moment the Forecast was approved by the IC to the cutoff date.

Czerka survival curve

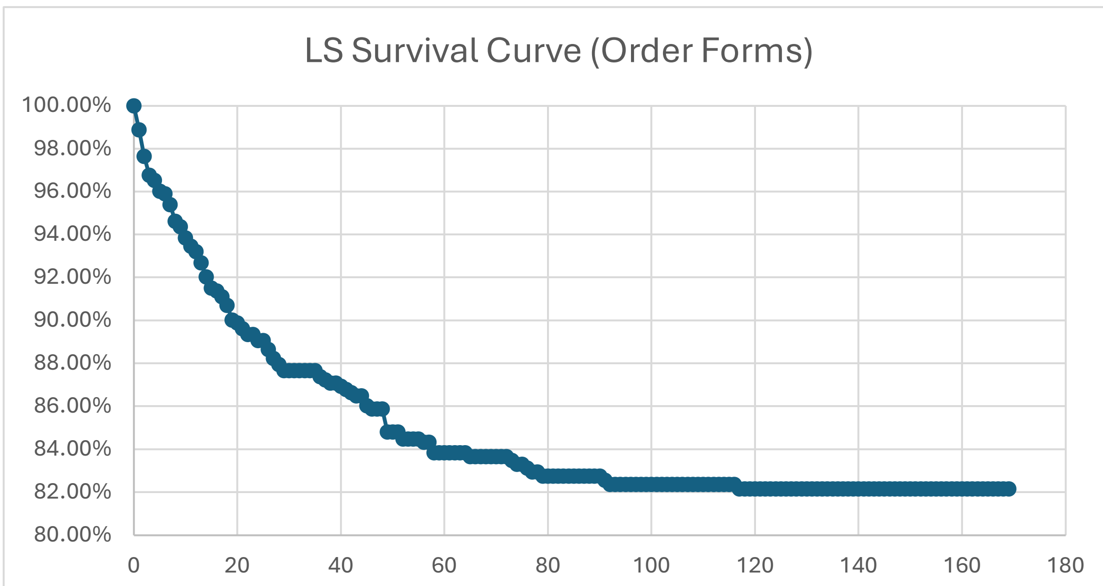
We pulled data from Czerka to extract the status of all Agreements and the dates they were last updated. After applying a simple cutoff threshold T_{95}, we found that 95% of all the deals that get signed are signed before 58.6 days have passed since the time of their creation. Any deal past 59 days is considered lost.

The resulting survival curve reveals an asymptotic plateau at a 30% cumulative conversion rate. This indicates that 70% of the total pipeline is effectively dead. Of the 30% of deals that do successfully execute, 95% achieve signature by day 59. Therefore, any open deal extending beyond day 59 belongs almost entirely to the stagnant 70%.



Because the unsigned deals outnumber the exceptional winners by hundreds to one, the marginal probability of any deal successfully signing after 59 days of existence is negligible (yielding a roughly 1% to 2% recovery rate).

However, Addenda inherently close faster than new Agreements because the baseline terms have already been negotiated. To prevent these from artificially shrinking our timeline, we isolated the survival curve strictly to standard Order Forms. Controlling for this variable shifts the T_{95} threshold to 73 days. This indicates that for genuinely new deals, 95% of successful signatures occur within this 73-day window.



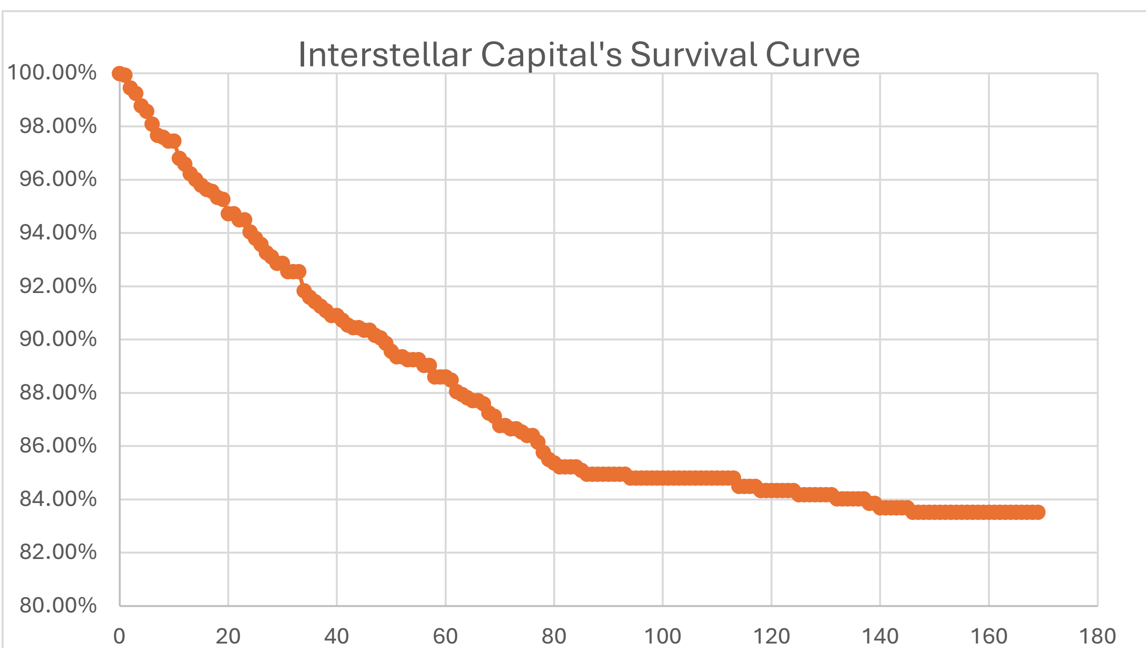
This filtered curve establishes a baseline: only 18% of Order Forms are successfully executed, and any standard deal past 74 days has a statistically insignificant probability of recovery. Is there a difference between existing and new partners?

Interstellar Capital survival curve

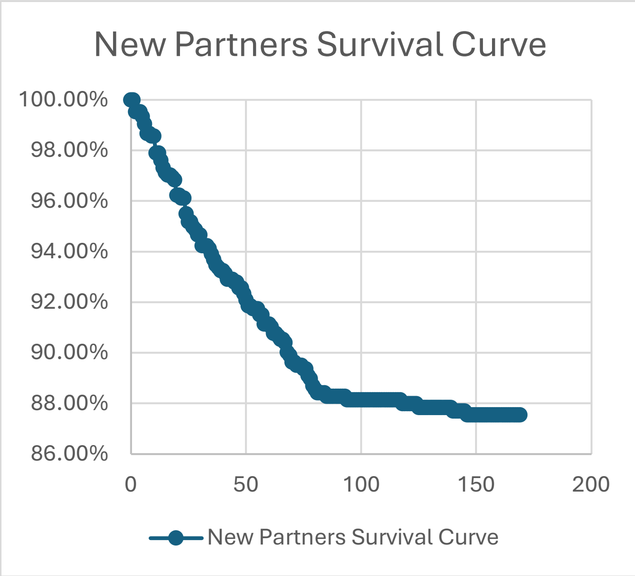
After matching the Interstellar Capital's (IC) approved deals to the signed deals found in Czerka, for the deals approved between November 1st, 2025, and until April 22nd, 2026, the raw conversion rate is 12.46%. That is, 561 deals out of 4,503 approved deals. Note that a minority of these are not unique, rather a Forecast resubmitted to the IC for approval. Furthermore, this assumption fails to account for sales cycles and cohort maturation. Since we know that many of those 4,503 deals are brand new and still actively negotiating, if we give those 4,503 deals enough time to finish their natural 170-day lifecycle, the final conversion rate will be higher.

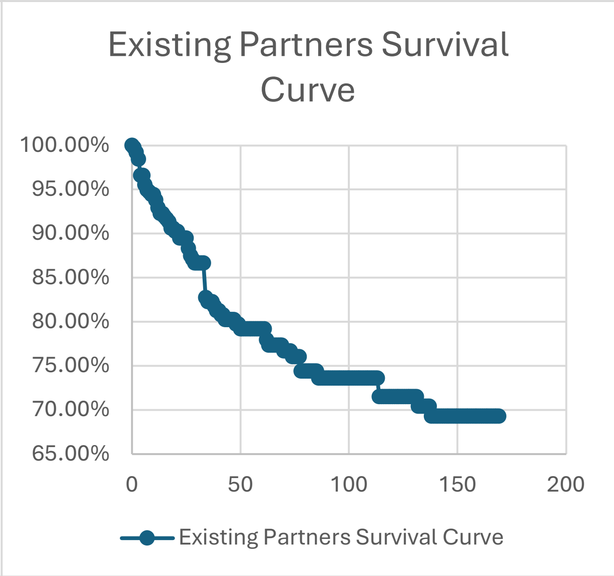
When preparing the survival curves, the fully signed dates were compared to the date a deal appeared for the first time in the selected period. Any deals that were signed, but came from the previous term, before November 1st, were discarded.

After applying the threshold T_{95}, we found that 95% of all the deals that get signed are signed before 163 days have passed since the time of their creation.



The survival curve shows that there is a cumulative conversion rate of 16%. Roughly 84% of the total pipeline does not get signed. Of the 16% of deals that do get signed, 95% of them achieve signature by day 163.





When comparing the difference between new and existing partners, the survival curves reveal that the conversion rate for new partners stops at 13%, while the existing partners curve shows that getting deals signed is significantly easier, with a signature rate of 30%. Both reach their threshold at roughly the same time as the aggregated survival curve. For the existing partners, the business model works and partners are willing to sign more units. However, new partners encounter an initial trust barrier.

Conclusion

What does the data show? It shows that the initial assumption that the rate of sales being below 10% is not accurate. The overall closing volume is near 30% for existing partners and 13% for new partners, which strongly aligns with existing industry data.

B2B service sales represent highly complex situations that require navigating numerous elements and stakeholders before a final deal can be reached. Data from CRM leaders and market analysts (e.g., HubSpot, Gartner) establishes the 90-to-180-day benchmark for mid-market and enterprise deals. This extended timeline is attributed to expanding buying committees, multi-stage risk mitigation, and the inherent structural complexity of B2B procurement.[4] CDH's lifecycle is within the standard 90-to-180-day mid-market benchmark.

By applying survival analysis, we have proven that once a deal is drafted, CDH's sales velocity and win rates are fundamentally healthy and it operates within industry standards both in its conversion rate and lifecycles. Further analysis could explore why nearly 50% of IC-approved deals are dying before a contract is ever generated in Czerka. The next critical step is to isolate the friction between IC approval and contract drafting and figuring out why potential partners lose interest in the program before a contract has been sent their way.

[4] "Sales Cycle Length Benchmark." Optifai. https://optif.ai/learn/questions/sales-cycle-length-benchmark/